

Do the information cost of a brain network model show a difference?

A research style project using Stochastic Linear, Diffusion Laplacian and Non-Linear Neural Mass Modelling comparison between shortest path ideal and empirical brain networks

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Introduction

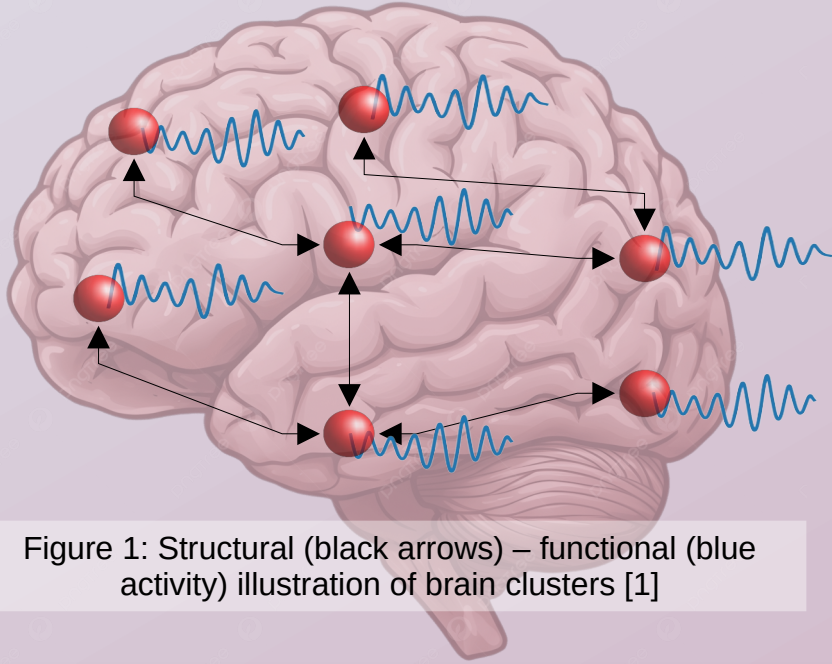


Figure 1: Structural (black arrows) – functional (blue activity) illustration of brain clusters [1]

Brain network modelling aims to describe the macroscopic behaviour of the brain by modelling interactions of clusters of neurons (neuron cluster regions, NCR). In this project, I model the network signalling dynamics of 246 brain regions using three different models.

The dataset consists of 100 participant. Their brains have been scanned when resting. The images are DTI and fMRI scans, from this the Structural Connectivity (SC) and Functional Connectivity (FC) is derived, as shown in Figure 2 for one of the individuals.

The central research question is whether brain communication is primarily based on the shortest-path routing. While shortest paths are efficient, they imply a high information cost, as real neuronal signalling is decentralised and lacks global knowledge of the network.

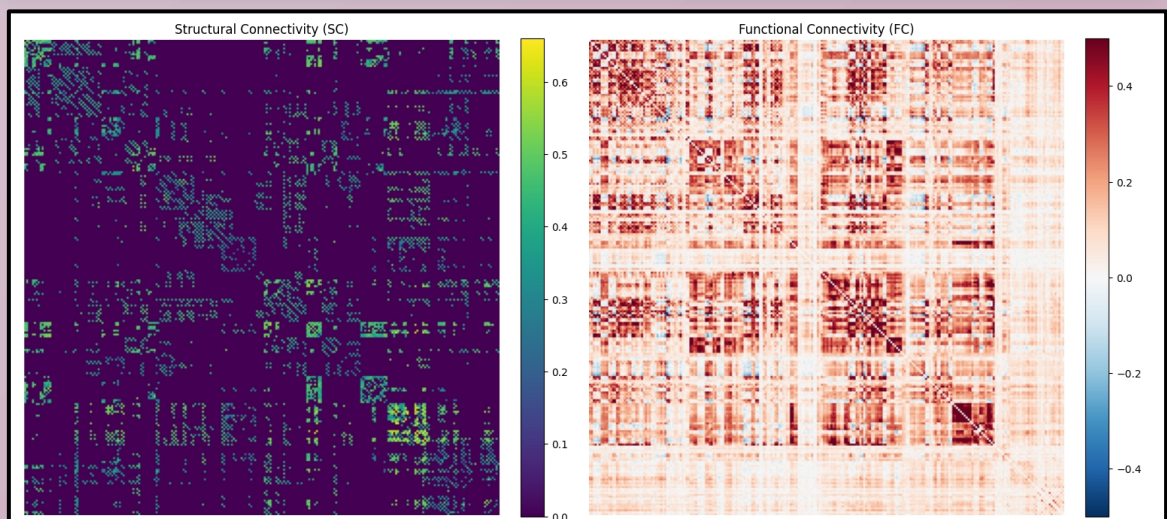


Figure 2: Showing SC and FC for 1 individual

Left: SC map which represents the white matter density (neuronal paths / axons) between regions. Symmetric, since large scale signalling assumed bidirectional
Right: FC map which represents linear correlations of signal activity of the regions. The diagonal has been removed for plot clarity (self connections).

Models

Three models were used [F. Abdelnour 2014]:

Stochastic Linear Model (SLM): This is a rather simple model that linearly relates the SC to the FC using two parameter and adds Gaussian noise with mean zero and standard deviation of one. The parameters were obtained by looping over the combination, choosing the one with the highest Pearson Correlation (PC) value.

Diffusion Laplacian Model (DLM): This model is also linear but calculates FC by diffusion behavior, where at $t=0 \rightarrow$ no correlation and $t=\infty \rightarrow$ everything correlated. Somewhere in between these extremes lies the actual network behaviour.

Non-Linear Neural Mass Model (NLNMM): This model is the most complex. It models the neural cluster behaviour based on biophysical behaviour of the neurons, with collective firing (total firing of the neurons in the group) and voltage levels modulated by calcium, sodium and potassium flows. The system is a 738 dimension dynamical system with 33 parameters.

The output of the models are then compared to empirical SC-FC, as shown in Figure 3.

Methodology

The models have been applied on the dataset. The SC was then rewired using a Shortest Path Ideal (SPI) algorithm.

The algorithm loops through the nodes one by one, finds all paths of length 2 that ends in the same terminal node, and then rewires the node's connections according to the shortest euclidean distance. The result of this algorithm is shown in figure 4 for one individual.

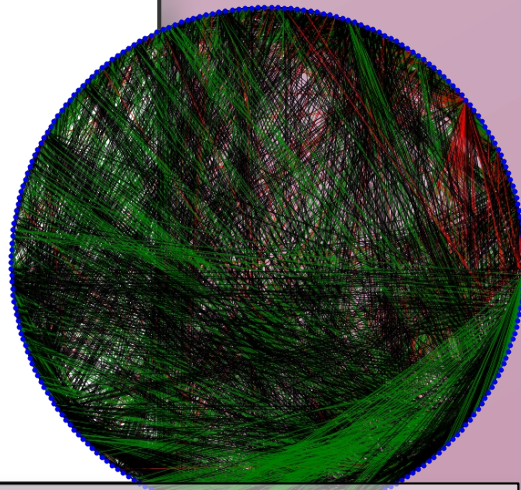


Figure 4: SPI Rewired SC plotted as a network graph

Blue dots are the NCR, Black is unchanged, Red is removed and green added connections

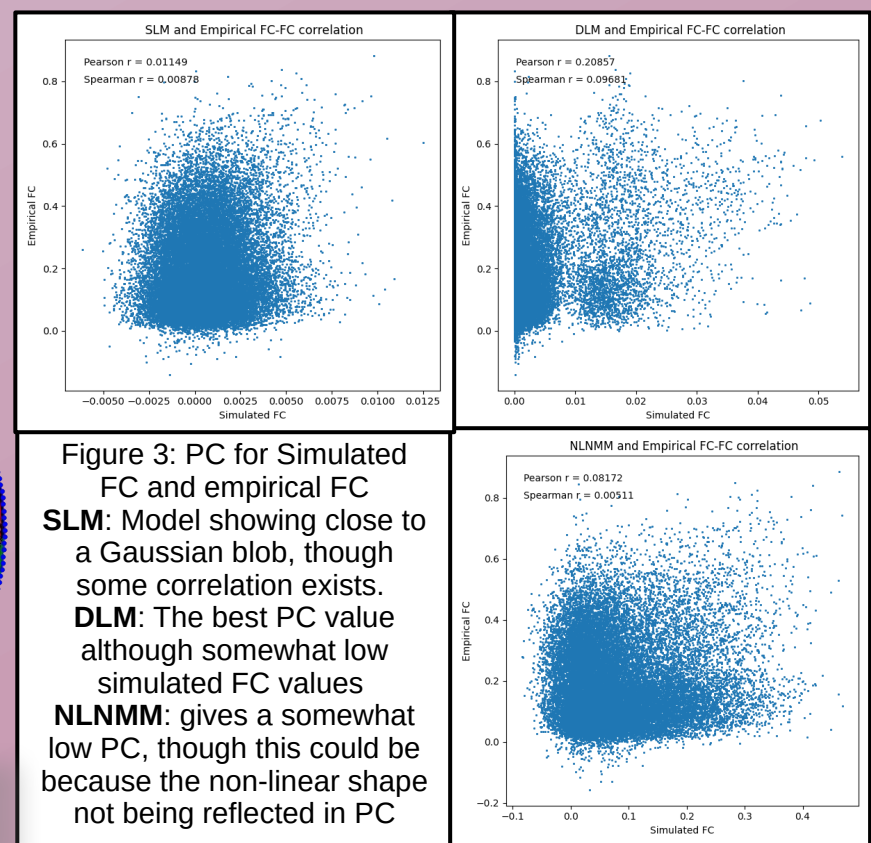


Figure 3: PC for Simulated FC and empirical FC

SLM: Model showing close to a Gaussian blob, though some correlation exists.

DLM: The best PC value although somewhat low simulated FC values

NLNMM: gives a somewhat low PC, though this could be because the non-linear shape not being reflected in PC

Results and discussion

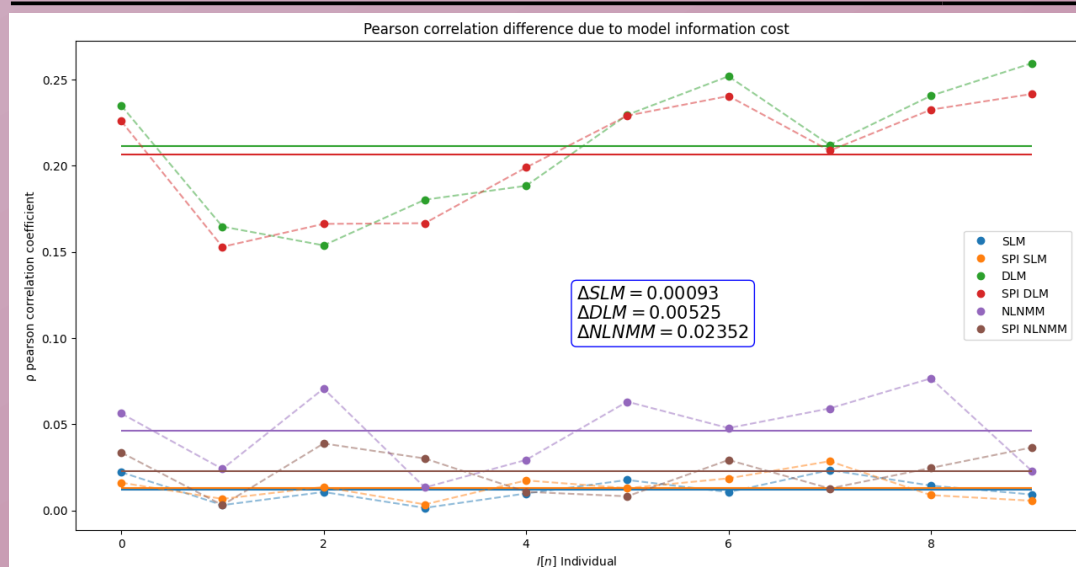


Figure 5: Results on FC-FC modelling Pearson correlation coefficient due to rewiring of SC

The SPI algorithm shows good results on the provided data. It finds shorter paths and rewires brain connections within the network. Without introducing large changes that would invalidate any SC-FC analysis. Across 10 individuals, the mean SP length (derived from SC network adjacency matrix) was reduced from ~ 2.575 to ~ 2.383 , which is a reduction by $\sim 7.5\%$.

However, the most important results are plotted in Figure 5. The outcomes of the SPI rewiring indicate that SLM, which has low information cost, is the least affected. In contrast, DLM and NLNMM both associated with higher information cost, are affected the most. Exhibiting reductions in accuracy by factors of $\sim 5,6$ and $\sim 25,3$ times.

This indicates that brain networks are not routed by shortest path and that this may affect model accuracy.